

Experiment-5

Interactive Dashboard Development using Tableau / Microsoft Power BI

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In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load Dataset
df = pd.read_csv(
    "Sample - Superstore.csv",
    encoding="latin1"
)
print("Dataset Loaded Successfully") print(df.head())

# Step 2: Check Dataset
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns.tolist())

# Step 3: Verify Data
print("\nData Types:")
print(df.dtypes)
print("\nMissing Values:")
print(df.isnull().sum())

# Step 4: Calculate KPIs
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
average_order_value = total_sales / total_orders
print("\n===== KPIs =====")
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Orders:", total_orders)
print("Average Order Value:", round(average_order_value, 2))

# Step 5: Bar Chart - Sales by Region
region_sales = df.groupby("Region")["Sales"].sum()
plt.figure(figsize=(8, 5))
region_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Step 6: Line Chart - Monthly Sales
df["Order Date"] = pd.to_datetime(df["Order Date"],
    format="mixed", errors="coerce")
monthly_sales = df.groupby(df["Order Date"].dt.to_period("M"))
```

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)["Sales"].sum()
plt.figure(figsize=(10, 5))
monthly_sales.plot(
kind="line",      marker="o"
)
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout() plt.show()

# Step 7: Pie Chart - Sales by Category
category_sales = df.groupby("Category")["Sales"].sum()
plt.figure(figsize=(7, 7)) category_sales.plot(
kind="pie",      autopct="%1.1f%%"
)
plt.title("Sales by Category")
plt.ylabel("")
plt.tight_layout() plt.show()

# Step 8: Correlation Heatmap numeric_data
= df.select_dtypes(include="number"
)
correlation = numeric_data.corr()
plt.figure(figsize=(9, 6))
sns.heatmap(      correlation,
annot=True,      fmt=".2f",
cmap="coolwarm"
)
plt.title("Correlation Heatmap")
plt.tight_layout() plt.show()

# Step 9: Filter / Slicer
print("\nAvailable Regions:")
print(df["Region"].unique())
region = input(
    "\nEnter Region (East/West/North/South): "
)
filtered_data = df[
df["Region"].str.lower() == region.lower()
]
print("\n===== Filtered Data =====")
print(filtered_data.head()) filtered_sales =
filtered_data["Sales"].sum() filtered_profit =
filtered_data["Profit"].sum() print("\nFiltered
Sales:",      round(filtered_sales, 2))
print("Filtered Profit:",
round(filtered_profit, 2))

# Step 10: Business Insights

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```
print("\n===== Business Insights =====")
top_region = region_sales.idxmax() top_category =
category_sales.idxmax() print("Top Sales Region:",
top_region) print(
    "Highest Region Sales:",
round(region_sales.max(), 2)
)
print("Top Sales Category:", top_category) print(
    "Highest Category Sales:",
round(category_sales.max(), 2) )
```

Dataset Loaded Successfully

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\ 0
1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520		
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	

	Customer Name	Segment	Country	City	...	\ 0
	Claire Gute	Consumer	United States	Henderson	...	
1	Claire Gute	Consumer	United States	Henderson	...	
2	Darrin Van Huff	Corporate	United States	Los Angeles	...	
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

	Postal Code	Region	Product ID	Category	Sub-Category	\ 0
42420	South	FUR-BO-10001798	Furniture	Bookcases		
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

	Product Name	Sales	Quantity	\ 0
	Bush Somerset Collection Bookcase	261.9600	2	
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400		
			3	
2	Self-Adhesive Address Labels for Typewriters b...	14.6200		
			2	
3	Bretford CR4500 Series Slim Rectangular Table	957.5775		
			5	
4	Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit	0
0.00	41.9136		
1	0.00	219.5820	
2	0.00	6.8714	
3	0.45	-383.0310	
4	0.20	2.5164	

[5 rows x 21 columns]

Dataset Shape:
(9994, 21)

Column Names:

['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit']

Data Types:

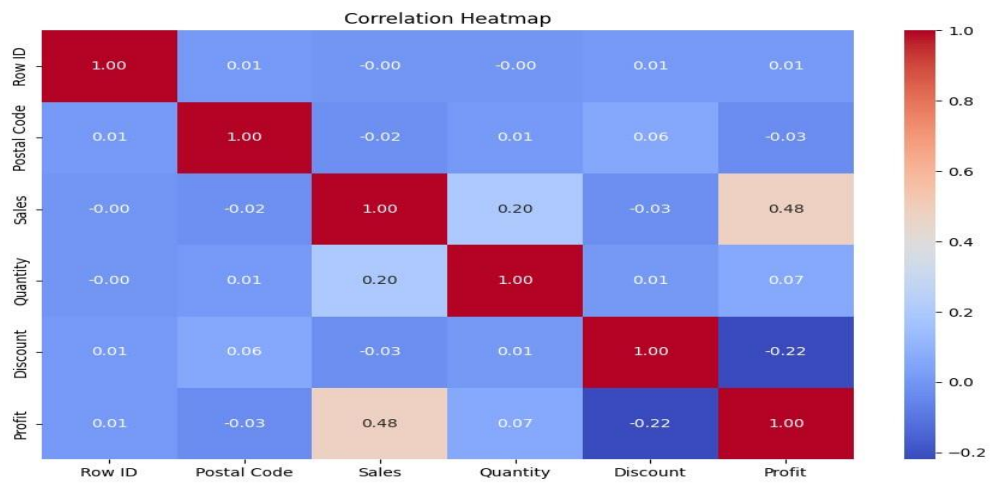
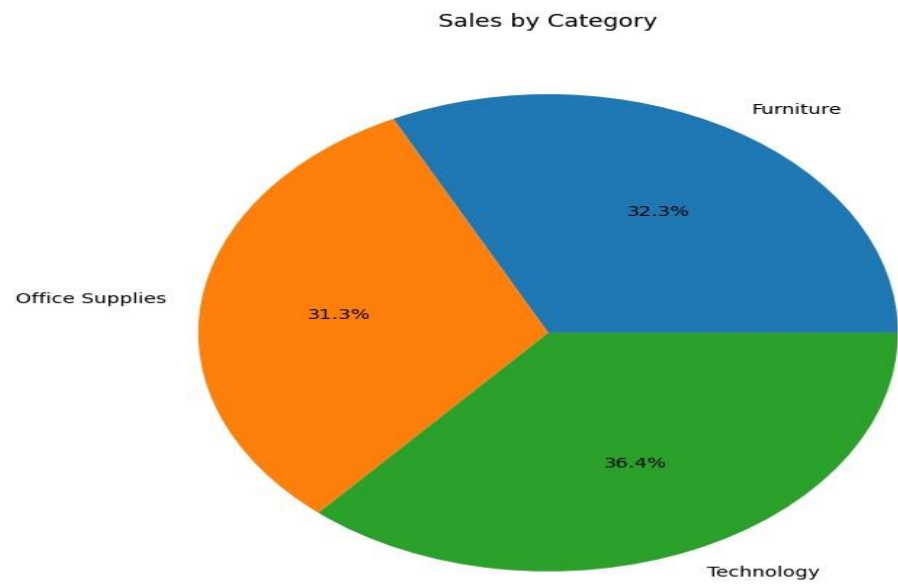
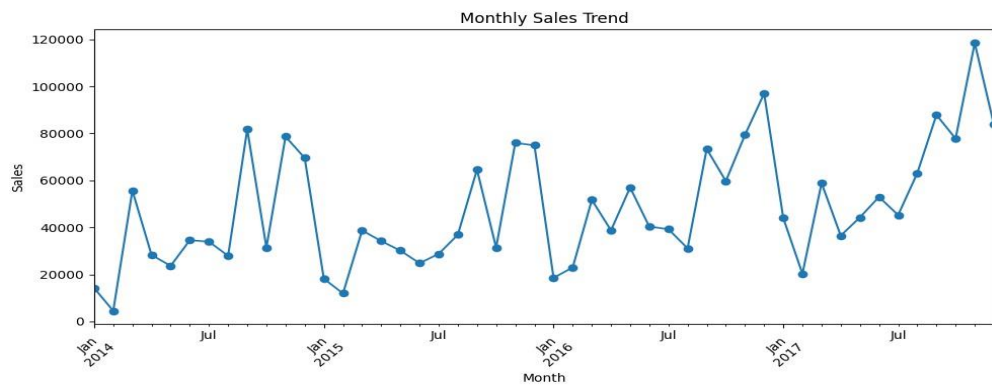
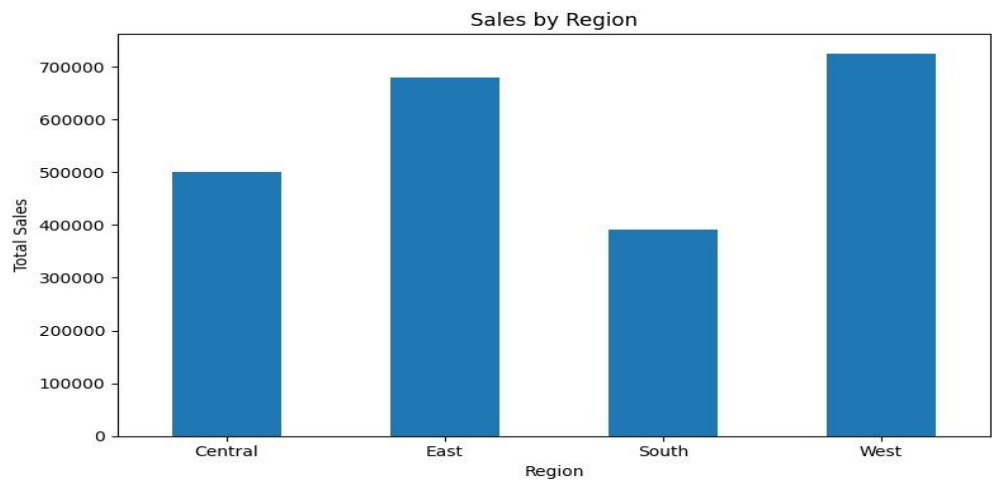
Row ID	int64
Order ID	object
Order Date	object
Ship Date	object
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
Country	object
City	object
State	object

```
Postal Code      int64
Region           object
Product ID       object
Category         object
Sub-Category     object
Product Name     object
Sales            float64
Quantity         int64
Discount         float64
Profit           float64
dtype: object
```

Missing Values:

```
Row ID          0
Order ID        0
Order Date      0
Ship Date       0
Ship Mode       0
Customer ID     0
Customer Name   0
Segment        0
Country         0
City            0
State           0
Postal Code     0
Region          0
Product ID      0
Category        0
Sub-Category    0
Product Name    0
Sales           0
Quantity        0
Discount        0
Profit          0
dtype: int64
```

```
===== KPIs =====
Total Sales: 2297200.86
Total Profit: 286397.02
Total Orders: 5009
Average Order Value: 458.61
```



Questions & Detailed Answers

1. What is Business Intelligence (BI)? How does it support organizational decision-making?

PDF

Business Intelligence (BI) is a collection of architectures, applications, and methodologies that gather, clean, integrate, and analyze operational data. It supports organizational decision-making by eliminating guesswork and transforming historical transactional logs into real-time KPI scorecards, interactive reports, and performance visualizations. This provides decision-makers with evidence-based insights to address revenue leaks, forecast demands, and streamline supply chains.

PDF+ 2

2. What is an interactive dashboard? How is it different from a static report?

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An interactive dashboard is a unified, visual display of key metrics and charts connected dynamically through a shared data model.

PDF

- **Static Report:** Fixed snapshot (e.g., standard PDF, paper report, or flat CSV) that conveys data in a rigid format without user modification.

PDF

- **Interactive Dashboard:** Empowers users to cross-filter across visuals, adjust dynamic slicers (by date, region, or category), and drill down through hierarchical levels to inspect underlying details directly on canvas.

PDF

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

PDF

Feature / Criteria	Microsoft Power BI PDF	Tableau PDF
Data Transformation & Modeling	Features Power Query (M language) for ETL and DAX for calculated relationship modeling. PDF	Requires cleaner source data upfront or the use of Tableau Prep Builder for complex ETL workflows. PDF
Visual Customization	Modular, drag-and-drop visuals with marketplace custom visual support. PDF	Highly detailed mark cards, dual axes, and fine-grained visual customization control. PDF
Usability & Learning Curve	Intuitive for users familiar with Microsoft Excel, tables, and Office 365 interfaces. PDF	Steeper initial learning curve, specifically when writing Level of Detail (LOD) expressions. PDF
Enterprise Application	Cost-effective, enterprise-scale reporting	Deep exploratory data discovery and complex visual storytelling,

Feature / Criteria	Microsoft Power BI PDF	Tableau PDF
	integrated within the Microsoft 365 and Azure ecosystems. PDF	commonly deployed in enterprise Salesforce/cloud stacks. PDF

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

PDF

KPIs are quantifiable indicators used to gauge an organization's performance against strategic objectives.

PDF

- **Total Sales / Revenue:** Gross transactional income over a selected duration.
PDF
- **Net Profit Margin:** Percentage of sales revenue remaining after all operating expenses and discounts.
PDF
- **Average Order Value (AOV):** Mean monetary amount spent per customer order.
PDF
- **Customer Retention / Churn Rate:** Proportion of repeat versus departed customers over a given interval.
- **Return on Investment (ROI) / Discount Impact:** Net profit relative to total promotional discounting applied.

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

PDF

- **Filters:** Backend or visual-level rules that exclude irrelevant records from the dataset (e.g., filtering out returned or canceled orders).
PDF
- **Slicers:** On-screen interactive controls (such as dropdowns, sliders, or button groups) that let end users isolate specific segments, time frames, or categories dynamically.
PDF
- **Drill-Down:** Enables hierarchical exploration within a single visual, allowing users to move from high-level aggregates down to granular roots (e.g., inspecting Annual Sales →Quarterly →Monthly, or Category →Sub-Category →Product ID).

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6. What factors should be considered while designing an effective business dashboard?

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- **Audience Persona:** Align detail level with user requirements—executive leadership needs high-level KPI cards, while logistics managers require granular tables.
PDF
- **Layout Hierarchy:** Place core KPIs at the top-left, progress charts in the center, and granular tables at the base.
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- **Clutter Reduction:** Remove redundant grid lines, non-essential 3D charts, and decorative visual noise.

- **Semantic Color Usage:** Use consistent color palettes (e.g., neutral slate tones for categories, red/green accents reserved strictly for loss/profit variances).
- **Responsive Layout:** Ensure dashboard layouts adapt legibly to desktop screens, tablets, and embedded web portals.

7. How can dashboards help managers monitor organizational performance in real time?

PDF

By connecting directly to live data warehouses or scheduled streaming feeds, dashboards update operational metrics continuously without manual reporting cycles. Managers can configure automated threshold triggers (such as inventory shortages, sudden revenue drops, or server spikes), enabling real-time detection of operational risks and faster intervention.

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8. Why is data visualization important in Business Intelligence tools?

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The human visual cortex processes visual representations such as size, position, and color substantially faster than raw alphanumeric rows. Visualizations highlight underlying trends, anomalies, clusters, and correlations that would remain unnoticed in massive relational tables, reducing cognitive load and accelerating decision cycles.

9. What insights can be derived from a sales dashboard for an e-commerce company?

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- **Product Profitability:** Pinpointing products that generate high sales volume but negative net margin due to excessive discounting.

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- **Regional Logistics Performance:** Identifying distribution zones suffering from shipment delays or high return rates.
- **Customer Segment Value:** Comparing average spend across consumer, corporate, and small-business buyer groups.
- **Seasonality Trends:** Identifying quarterly or month-end purchasing spikes to optimize supply chain inventory.

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10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

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Interactive dashboards establish a unified, transparent source of truth that bridges technical data models and executive decision-making. Instead of relying on static presentation decks that require subsequent follow-up queries, stakeholders can test assumptions and validate ad-hoc questions live during meetings by toggling slicers and filters, focusing discussions on strategy rather than report generation.